

Rabbit App: One-Page Summary

Repo evidence only

What It Is

Rabbit is a full-stack e-commerce web app with a React frontend and an Express plus MongoDB backend. It supports customer shopping flows and admin operations from the same codebase.

Who It's For

Primary persona: a small e-commerce team that needs both a customer storefront and an admin panel for managing catalog, users, and orders.

What It Does

- User authentication with JWT login and registration, plus protected admin routes.
- Product catalog browsing with collection, category, price, color, brand, material, and search filters.
- Guest and signed-in carts, including guest-to-user cart merge on login.
- Checkout workflow with payment status tracking and order finalization.
- PayPal button integration in the frontend checkout flow.
- Admin CRUD for users, product management, and order status updates.
- Image upload endpoint that streams files to Cloudinary; newsletter subscription endpoint.

How It Works (Architecture)

- Frontend: React 19 + Vite + React Router + Redux Toolkit state slices (auth/products/cart/checkout/orders/admin).
- Backend API: Express app mounts /api/users, /api/products, /api/cart, /api/checkout, /api/orders, /api/upload, /api/subscribe, and admin routes.
- Data layer: Mongoose models (User, Product, Cart, Checkout, Order, Subscriber) connected via MONGO_URI.
- Security and flow: JWT bearer token verified in middleware; frontend stores token in localStorage and sends it on protected requests.
- Payments and media: PayPal client in frontend; Cloudinary upload route in backend for product images.
- Not found in repo: queue/message broker, caching layer, API gateway, or formal system diagram.

How To Run (Minimal)

- Install dependencies: run npm install in project root, backend/, and frontend/.
- Set environment files: backend/.env needs PORT, MONGO_URI, JWT_SECRET, CLOUDINARY_*; frontend/.env needs VITE_BACKEND_URL and VITE_PAYPAL_CLIENT_ID.
- Start backend: from backend/, run npm run dev.
- Start frontend: from frontend/, run npm run dev.
- Open the Vite local URL and ensure VITE_BACKEND_URL targets the backend port.